SIEMENS MANUFACTURING OEE SQL ANALYSIS

1. What records are in the table?

date	machine_id	shift	planned_hours	actual_hours	availability_pct	performance_pct	quality_pct	oee_pct	units_produced	defects
2023-07-28	MCH-004	Night	8.00	7.34	0.91700	0.75400	0.88100	0.60900	4040	480
2024-03-29	MCH-004	Night	8.00	5.99	0.74900	0.80500	0.96700	0.58300	3107	102
2024-04-01	MCH-002	Morning	8.00	7.53	0.94100	0.77600	0.89200	0.65100	3211	346
2023-01-01	MCH-005	Afternoon	8.00	6.61	0.82600	0.95100	0.99200	0.77900	4806	38
2023-02-24	MCH-018	Night	8.00	7.86	0.98200	0.71200	0.98300	0.68700	3892	66
2023-06-29	MCH-009	Morning	8.00	6.79	0.84900	0.97600	0.99100	0.82100	4823	43
2023-12-03	MCH-006	Night	6.00	4.54	0.75600	0.98000	0.93900	0.69600	4077	248
2023-02-25	MCH-013	Morning	8.00	6.28	0.78500	0.87100	0.90700	0.62000	4134	384
2023-09-28	MCH-011	Afternoon	8.00	6.71	0.83900	0.88500	0.91900	0.68200	4532	367
2023-01-29	MCH-013	Night	8.00	6.48	0.81000	0.92200	0.88900	0.66400	4653	516

2. How many machines, shifts, and dates are represented?

machine_count	shift_count	first_date	last_date
20	3	2023-01-01	2024-05-14

3. What is the average OEE?

average_oee
0.643695071

4. What are the average availability, performance, and quality values?

average_avalability_pct	average_performance_pct	average_quality_pct
0.817032854	0.839510266	0.938840862

5. How many units and defects are in the data, and what is the defect rate?

total_units	total_defects	defect_rate
3722340	228276	6.1326

6. Which shift has the highest average OEE?

shift	average_oee_pct
Morning	0.645559270
Afternoon	0.642875399
Night	0.642620481

7. Which machines have the highest and lowest average OEE?

machine_id	average_oee_pct
MCH-013	0.598409090
MCH-020	0.669977777

8. How does OEE vary by month?

yr	mth	average_pct
2023	1	65.8
2023	2	65.1
2023	3	66.1
2023	4	62.4
2023	5	65.1
2023	6	65.2
2023	7	63.2
2023	8	65.6
2023	9	63.6
2023	10	64.9
2023	11	65.2
2023	12	62.2
2024	1	63.1
2024	2	64.3
2024	3	64.6
2024	4	65.6
2024	5	60.8

9. Which shift has the highest and lowest average OEE ?

shift	average_pct
Morning	64.6
Night	64.3
Afternoon	64.3

10. Which shift has the highest defect rate?

shift	defect_rate
Night	6.3201
Afternoon	6.2587
Morning	5.8257

11. Which machines produced the most units?

machine_id	total_units
MCH-002	220326
MCH-018	217569
MCH-003	214191
MCH-004	203217
MCH-012	198570

12. In the lowest quarter of OEE records, which component is weakest?

record_count	average_oee	average_availability	average_performance	average_quality
244	0.521045081	0.720889344	0.778143442	0.932229508

13. What is each machine's worst OEE record?

date	machine_id	shift	oee_pct
2023-01-11	MCH-015	Afternoon	0.41700
2023-04-14	MCH-004	Morning	0.43000
2023-10-16	MCH-007	Night	0.43000
2023-10-25	MCH-018	Night	0.43200
2023-08-13	MCH-010	Afternoon	0.43600
2023-12-18	MCH-014	Morning	0.43700
2023-06-24	MCH-012	Night	0.43800
2024-01-19	MCH-008	Afternoon	0.43900
2023-03-01	MCH-009	Afternoon	0.43900
2024-01-23	MCH-006	Afternoon	0.44000
2023-11-08	MCH-013	Afternoon	0.45100
2023-03-13	MCH-005	Afternoon	0.45300
2023-09-15	MCH-016	Morning	0.45800
2023-04-22	MCH-001	Morning	0.45900
2023-01-04	MCH-011	Morning	0.46100
2023-03-18	MCH-019	Afternoon	0.47400
2023-12-29	MCH-020	Afternoon	0.47800
2023-05-14	MCH-003	Afternoon	0.47900
2023-08-19	MCH-002	Morning	0.48100
2023-12-09	MCH-017	Night	0.49900

14. How did monthly OEE change compared with the previous month?

month_start	average_oee	previous_month_oee	change_from_previous_month
2023-01-01	65.8	NULL	NULL
2023-02-01	65.1	65.8	-0.7
2023-03-01	66.1	65.1	1.0
2023-04-01	62.4	66.1	-3.7
2023-05-01	65.1	62.4	2.7
2023-06-01	65.2	65.1	0.1
2023-07-01	63.2	65.2	-2.0
2023-08-01	65.6	63.2	2.4
2023-09-01	63.6	65.6	-2.0
2023-10-01	64.9	63.6	1.3
2023-11-01	65.2	64.9	0.3
2023-12-01	62.2	65.2	-3.0
2024-01-01	63.1	62.2	0.9
2024-02-01	64.3	63.1	1.2
2024-03-01	64.6	64.3	0.3
2024-04-01	65.6	64.6	1.0
2024-05-01	60.8	65.6	-4.8

15. Which machines have the highest average OEE within each shift?

shift	machine_id	record_count	average_oee
Afternoon	MCH-014	17	0.696588235
Morning	MCH-006	14	0.693714285
Night	MCH-016	20	0.686300000

16. What is the running total of units produced over time?

date	units_for_day	running_units
2023-01-01	9253	9253
2023-01-02	13535	22788
2023-01-04	4023	26811
2023-01-06	8439	35250
2023-01-07	13557	48807
2023-01-08	11052	59859
2023-01-09	12355	72214
2023-01-11	17840	90054
2023-01-12	9162	99216
2023-01-14	8105	107321
2023-01-15	11831	119152
2023-01-16	16271	135423
2023-01-18	5858	141281
2023-01-19	8336	149617
2023-01-20	8604	158221
2023-01-21	8443	166664
2023-01-22	16816	183480
2023-01-23	10151	193631
2023-01-24	7413	201044
2023-01-25	11112	212156
2023-01-26	7618	219774
2023-01-27	7850	227624
2023-01-28	6355	233979